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Osteoarthritis

# Deep learning–based clustering for endotyping and post-arthroplasty response classification using knee osteoarthritis multiomic data

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ABSTRACT

*Objectives:* Primary knee osteoarthritis (KOA) is a heterogeneous disease with clinical and molecular contributors. Biofluids contain microRNAs and metabolites that can be measured by omic technologies. Multimodal deep learning is adept at uncovering complex relationships within multidomain data. We developed a novel multimodal deep learning framework for clustering of multiomic data from 3 subject-matched biofluids to identify distinct KOA endotypes and classify 1-year post–total knee arthroplasty (TKA) pain/function responses.

*Methods:* In 414 patients with KOA, subject-matched plasma, synovial fluid, and urine were analysed using microRNA sequencing or metabolomics. Integrating 4 high-dimensional datasets comprising metabolites from plasma and microRNAs from plasma, synovial fluid, or urine, a multimodal deep learning variational autoencoder architecture with K-means clustering was employed. Features influencing cluster assignment were identified and pathway analyses conducted. An integrative machine learning framework combining 4 molecular domains and a

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clinical domain was then used to classify Western Ontario and McMaster Universities Arthritis Index (WOMAC) pain/function responses after TKA within each cluster.
*Results:* Multimodal deep learning−based clustering of subjects across 4 domains yielded 3 distinct patient clusters. Feature signatures comprising microRNAs and metabolites across biofluids included 30, 16, and 24 features associated with clusters 1 to 3, respectively. Pathway analyses revealed distinct pathways associated with each cluster. Integration of 4 multiomic domains along with clinical data improved response classification performance, surpassing individual domain classifications alone.
*Conclusions:* We developed a multimodal deep learning−based clustering model capable of integrating complex multifluid, multiomic data to assist in uncovering biologically distinct patient endotypes and enhance outcome classifications to TKA surgery, which may aid in future precision medicine approaches.

### WHAT IS ALREADY KNOWN ON THIS TOPIC

- Searching the PubMed database for articles published up to October 7, 2024, using the search string "osteoarthritis AND endotype AND (classify OR classification OR prediction OR predictive) AND (machine learning OR deep learning) NOT review" only uncovered 3 articles, with 2 relevant articles using machine learning approaches to identify endotypes of knee or hip osteoarthritis (OA) patients.
- With omic technologies capable of evaluating thousands of features from multiple disease-relevant biofluids, more advanced modelling techniques able to integrate multiple omic feature sets are needed.

### WHAT THIS STUDY ADDS

- This study uses a large sample of patients with knee osteoarthritis (KOA) containing multiomic (miRNomics and metabolomics) molecular data from diverse patient-matched biofluids (synovial fluid, plasma and urine).
- We developed a novel multimodal deep learning and machine learning framework to seamlessly integrate these high-dimensional, multimodal data to enhance the understanding of KOA by identifying distinct patient clusters and improve classification of postsurgical pain and function responses.
- Our integrated approach goes beyond the capabilities of individual domains, providing a more comprehensive methodological approach to understand disease and treatment outcomes.
- The entire sample dataset, including multiomic measures and modelling framework, are publicly available.

### HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- Our methodology may assist in uncovering multiple clusters of patient endotypes with unique physiologically relevant pathways, which can be evaluated in outcome classification testing, such as for pain and function responses to total knee arthroplasty (TKA) surgery.
- Integration of additional omic technology data and clinical measures will be crucial to refine endotype clustering and response classifications using this modelling approach.
- Our approach may ultimately aid joint patient−physician decision making for future precision medicine therapeutic approaches, as well as improve subject selection criteria for future clinical trials.

## INTRODUCTION

Osteoarthritis (OA) is a degenerative, painful, and disabling joint disease affecting over 500 million people worldwide [1], with the knee most commonly afflicted [2]. Primary knee OA (KOA) patients are heterogeneous [3]. Risk factors of KOA include age, sex (defined as patient-reported male/female), and obesity status [2]. Mental health and persistent pain status have also been associated with KOA clinical phenotypes [4,5]. Total joint arthroplasty (TKA) is the only available therapy for patients with KOA who no longer respond to conservative management; however, up to 34% of patients with KOA that undergo TKA fail to achieve clinically relevant pain reduction [6]. Identifying those at high risk of nonresponse is of significant interest. Psychosocial and sociodemographic variables are likely insufficient to explain differences in patient outcomes to TKA [7−9]. However, it is possible that KOA heterogeneity captured by biological features may improve our ability to classify patient responses to TKA.

Biofluid microRNAs (miRNAs) and metabolites can provide highly descriptive, individualised categorisations of patients beyond clinical measures. MiRNAs epigenetically modify target RNA expression. Biofluid metabolomes represents snapshots of the metabolic activity contributed by associated cells and tissues. Advanced omic technologies can measure miRNAs (miRNomics) and metabolites (metabolomics), primarily by next-generation sequencing (NGS) and liquid chromatography-mass spectrometry/mass spectrometry (LC-MS/MS), respectively [10,11].

In a case-control study using the UK Biobank cohort, 14 distinct OA risk phenotypes were identified by multimodal machine learning (ML) using clinical factors alone [12]. The inclusion of individual proteomic, genomic or metabolomic data showed no prediction improvement of case-control status over clinical factor modelling alone [12]. In contrast, studies using individual biofluids have identified endotypes of patients with OA. Three endotypes of patients with OA (low tissue turnover, structural damage and systemic inflammation) were identified from a panel of 16 serum and urine proteins and peptides using unsupervised ML [13]. Plasma metabolomics alone uncovered multiple endotypes of patients with KOA [14,15], with some metabolite ratios able to differentiate specific endotypes of KOA subjects from control participants [14]. KOA patient biofluid miRNA signatures were also able to differentiate between slow and fast progressors [16], early and late KOA [17,18], and patients requiring TKA or not [19]. Thus, endotype data from biofluids are important for understanding KOA heterogeneity, and consequently, may be associated with patient outcomes. To our knowledge, KOA endotypes have yet to be evaluated across multiple biofluids using multiomic technologies in an integrated approach [11].

Our experience to date suggests that multi-biofluid, multiomic endotyping requires more complex modelling systems. Deep learning aids in extracting complex patterns from data.

Autoencoders are one such deep learning approach. Although standard autoencoders focus on reconstructing input data by mapping them to and from a compressed latent representation without imposing a probabilistic structure, variational autoencoders (VAEs) introduce a probabilistic framework by parameterising the latent space using distributions, such as Gaussian, rather than deterministic points. This allows VAEs to incorporate stochastic sampling during training, capturing inherent data variability and improving robustness, key attributes for clustering in high-dimensional, multimodal omic data. Integrating multimodal data from multiple sources presents challenges due to the diversity within and across data types. VAEs address this challenge by embedding diverse data domains into reduced latent dimensions, facilitating improved data clustering [20−22]. Despite the potential of VAEs, there is a lack of unified frameworks for leveraging these methods to identify clusters from multimodal data and to classify clinical responses by integrating diverse data domains. Additionally, VAEs have not been applied to investigate OA endotypes.

In this study, we developed a novel multimodal deep learning framework employing VAEs for integrative clustering using 4 high-dimensional domains of subject-matched multiomic data from synovial fluid, urine and plasma, and tested its ability to determine distinct clusters (endotypes) of a sample of patients with KOA. Leveraging these endotypes, we further developed an integrative ML framework and tested the potential of this methodology to assess pain and function responses to TKA surgery.

## METHODS

### Study sample

A sample of 414 patients with primary KOA who underwent TKA within the Longitudinal Evaluation in the Arthritis Program-OA cohort (University Health Network, Toronto, Ontario, Canada), as previously described [23], and who had synovial fluid (collected intraoperatively), plasma and urine (collected up to 3 months prior to surgery) available, were selected for analysis. Nonfasting blood was collected for plasma, and urine was collected at first miction. All subjects were radiographic Kellgren−Lawrence (KL) grade III/IV primary KOA, were symptomatic, and fulfilled the American College of Rheumatology clinical criteria for KOA classification [24]. Subjects completed self-reported, multidimensional questionnaires from which baseline Western Ontario and McMaster Universities Arthritis Index (WOMAC) pain and function scores were calculated, with a maximum 20 points for pain and 68 points for function, with higher scores indicative of greater pain or more disability, respectively [25]. Hospital Anxiety and Depression Scale (HADS) [26] and painDETECT [27] scores at baseline (completed within the 3 months preceding surgery), and WOMAC pain and function scores at 1 year after TKA were also calculated. Improvement in WOMAC pain and function from baseline to 1 year after TKA was calculated, and individuals were categorised as responders (>33% improvement) or nonresponders (≤33% improvement), consistent with moderately important pain change thresholds [28,29], and used in a previous report [23]. HADS depression and anxiety scores were each categorised as normal (score 0-7), borderline case (score 8-11), or definite case (score 11-21) [26]. PainDETECT neuropathic-like pain scores were used to classify subject pain as likely nociceptive (score 0-12), unclear (score 13-18), or likely neuropathic-like (score 19-38) [27]. Baseline age, sex, and weight and height (from which body mass index [BMI; kg/m$^2$] was calculated) were also collected. Biofluids and patient data were collected with written informed consent from each patient and Research Ethics Board approval of the University Health Network, Toronto, Ontario (REB# 07-0383-BE; 14-7592-AE).

### MiRNA sequencing (miRNomics) and metabolomics

MiRNA extraction was performed using 200 µL plasma (N = 414), 100 µL synovial fluid (N = 414), and 1 mL urine (N = 414). Complementary DNA (cDNA) libraries were prepared using a protocol we previously reported [18]. NGS was conducted at the Schroeder Arthritis Institute (University Health Network, Toronto, Ontario) sequencing facility using the Illumina NextSeq550 platform. Alignment, processing, and quality assessment were performed using a previously reported pipeline [30]. Targeted metabolomics was used to profile 188 metabolites (Biocrates AbsoluteIDQ p180 kit, Biocrates Life Sciences AG) in N = 414 plasma samples at The Metabolomics Innovation Centre (Calgary, Alberta, Canada) by LC-MS/MS, as previously described [15]. Metabolite quantification and batch correction were conducted using the Absolute IDQ-coupled MetIDQ software (Biocrates). MiRNA count data and metabolite concentrations were normalised using sum normalisation, log-transformation, and Pareto scaling [31]. To stabilise variance estimates within differential expression (DE) analysis, empirical Bayes moderation techniques were applied.

### OmicVAE: integrative VAE architecture for multimodal clustering

We generated a novel VAE architecture named ‘*omicVAE*’ designed to cluster multimodal multiomic data (Fig 1). *OmicVAE* consists of a single encoder network followed by 4 individual decoder networks to perform integrative clustering combining 4 modalities: plasma metabolites, plasma miRNAs, synovial fluid miRNAs, and urine miRNAs.

The encoder network inputs concatenated multimodal multiomic data and maps them to a shared latent space representation using multiple fully connected neural network (FNN) layers with a sigmoid activation function. The encoder network’s output layers parameterise the mean and variance of a Gaussian distribution representing the shared latent space. Each decoder network reconstructs its respective domain’s input data from samples drawn from this latent space, using multiple FNN layers. During training, variational inference optimises the VAE’s parameters. The objective function $L_{total}$ includes the reconstruction loss ($L_{rec}$) and the Kullback−Leibler divergence ($L_{KL}$) between the learned latent distribution and a predefined prior (Equations 1-3). Minimising Kullback-Leibler divergence regularises the latent space, preventing overfitting and ensuring it remains structured and interpretable. The reconstruction loss measures the discrepancy between the input data and its reconstruction by the VAE decoder for each modality.

$$\begin{aligned} L_{rec} = & \frac{1}{N_{metabolites}} \sum_{i=1}^{N} \left\| X_{metabolites(i)} - Decoder_{metabolites(i)}(z) \right\|^2 \\ & + \frac{1}{N_{miRNA_{plasma}}} \sum_{i=1}^{N} \left\| X_{miRNA_{plasma}(i)} - Decoder_{miRNA_{plasma}(i)}(z) \right\|^2 \\ & + \frac{1}{N_{miRNA_{synovial}}} \sum_{i=1}^{N} \left\| X_{miRNA_{synovial}(i)} - Decoder_{miRNA_{synovial}(i)}(z) \right\|^2 \\ & + \frac{1}{N_{miRNA_{urine}}} \sum_{i=1}^{N} \left\| X_{miRNA_{urine}(i)} - Decoder_{miRNA_{urine}(i)}(z) \right\|^2 \end{aligned} \quad (1)$$

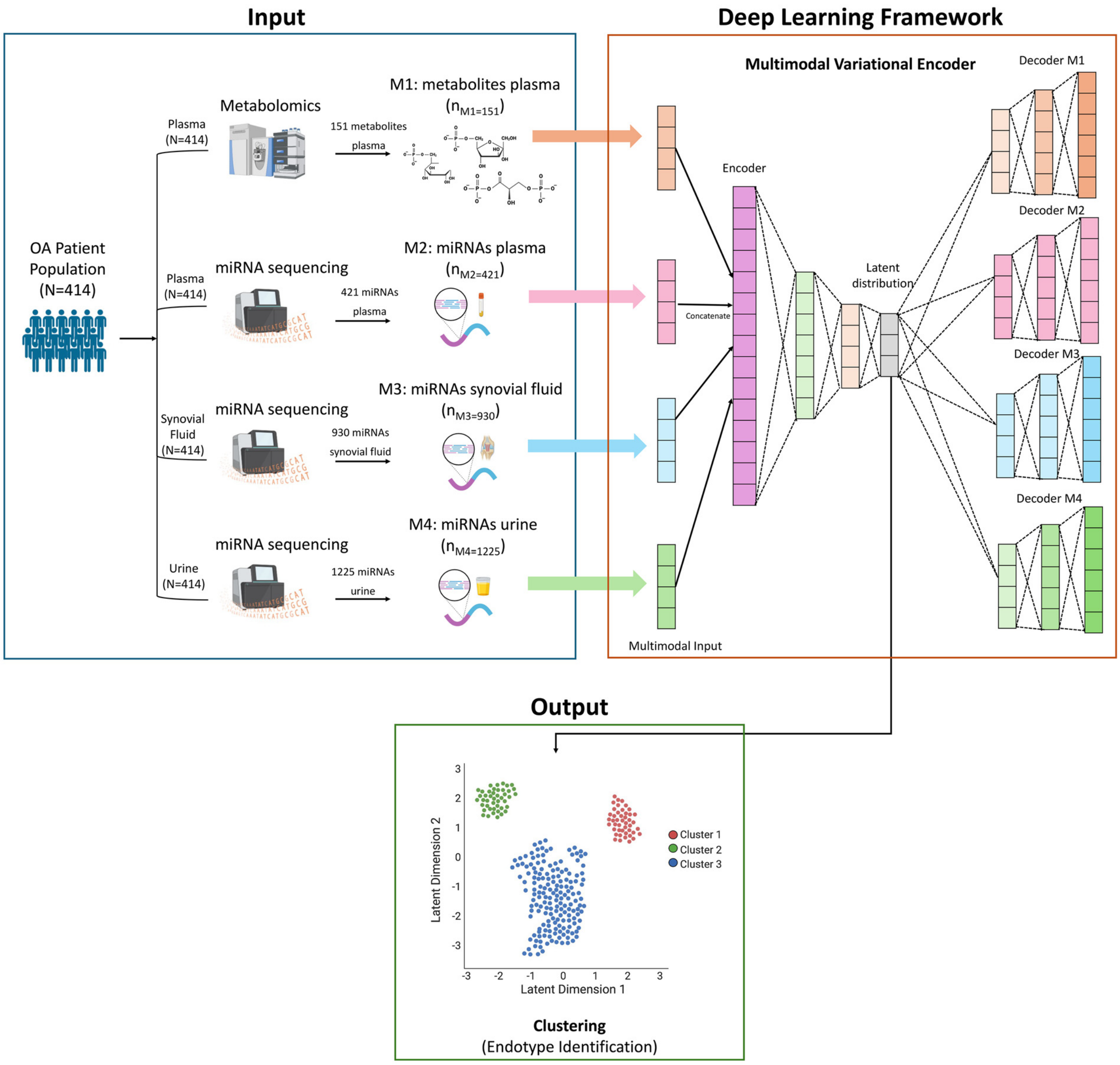


**Figure 1. Overall framework for deep learning−based multimodal clustering.** Deep learning framework for multimodal integration and clustering using variational autoencoder (VAE) modelling. Four individual decoders in the proposed VAE accurately identify the latent distribution within each domain (M1-M4) capturing the complex nonlinear associations within the multimodal data. K-means approach is used for identifying 3 clusters from the obtained latent distribution. 'N' represents the number of patients, and '$n_{Mx}$' represents the number of features in each data domain. Created, in part, with BioRender.com. OA, osteoarthritis; miRNA, microRNA.

$$
\begin{aligned}
L_{KL} = & -\frac{1}{2}\Bigg(\left(1+\log\left(\sigma^2_{metabolites}\right)-\mu^2_{metabolites}-\sigma^2_{metabolites}\right)\\
& +\left(1+\log\left(\sigma^2_{miRNA_{plasma}}\right)-\mu^2_{miRNA_{plasma}}-\sigma^2_{miRNA_{plasma}}\right)\\
& +\left(1+\log\left(\sigma^2_{miRNA_{synovial}}\right)-\mu^2_{miRNA_{synovial}}-\sigma^2_{miRNA_{synovial}}\right)\\
& +\left(1+\log\left(\sigma^2_{miRNA_{urine}}\right)-\mu^2_{miRNA_{urine}}-\sigma^2_{miRNA_{urine}}\right)\Bigg)
\end{aligned} \tag{2}
$$

$$
L_{total} = L_{rec} + L_{KL} \tag{3}
$$

In Equations 1 to 3, i represents the samples in each modality, X is the input data, Decoder(z) is the reconstructed data, and $\mu$ and $\sigma$ denote the mean and variance of the Gaussian distribution in the latent space, respectively. Once omicVAE is trained, K-means clustering on the learned latent space is used to identify distinct subpopulations within the multimodal multiomic data.

### *Multiomic signature identification within each cluster*

We employed a comprehensive approach to identify signature features (miRNAs and/or metabolites within 3 biofluids) influencing cluster assignment within each domain. We concurrently conducted standardised mean differences (SMDs) and DE analyses for pairwise cluster comparisons (one cluster vs others), using Benjamini−Hochberg (BH) adjusted *P* values (q < 0.05) to identify statistically different features. By integrating these analyses, we identified features with both large SMDs and significant DE, capturing robust signature features distinguishing clusters.

### *Endotype pathway analysis*

Gene targets of miRNAs per cluster were identified using the top 1% of targets per miRNA using mirDIP 5.2 (https://ophid.utoronto.ca/mirDIP) [32]. We performed pathway enrichment analysis for sets of gene targets in each cluster using pathDIP 5 (https://ophid.utoronto.ca/pathDIP) [33]. Diseases, drugs and vitamins, and genetic information processing pathway types were excluded from enrichment analysis. Only pathways with q value (BH adjusted) <0.01 were considered. Metabolite pathway enrichment analysis was not possible, so we identified pathways that included metabolites specific for each cluster for further analyses. Selected pathways specific for each cluster were visualised using NAViGaTOR 3.0.19 (https://navigator.ophid.utoronto.ca/navigatorwp) [34]. Mapping of pathways to consolidated categories in pathDIP was used to calculate the number of pathways per category. ggradar2_1.1.0 in R 4.3.0 was subsequently used to plot their distribution per cluster, scaling category pathway counts from 0% to 100%.

### *Integrative ML framework for classifying response*

We developed a comprehensive 2-step ML framework to integrate plasma metabolite, plasma miRNA, synovial fluid miRNAs and urine miRNA domains, along with a clinical domain (consisting of age, sex, BMI, depression and anxiety categories, and neuropathic-like pain category), to classify 1-year pain and function responses (i.e., responders vs nonresponders). In the first step, we trained separate unimodal ML models for each domain to extract features classifying 1-year response. We used the *mice* (multivariate imputation by chained equations) package (version 3.14.0) in R (R Foundation for Statistical Computing, Vienna, Austria) to perform multivariate imputation by chained equations of the missing clinical data (missingness <8%). We explored various ML algorithms including logistic regression, lasso regression, ridge regression, support vector machines and random forests, selecting models based on 10-fold cross-validation performance.

In the second step, we integrated features from all domains using a Naïve Bayes meta-classifier, trained with classifiers from the unimodal models. Cross-validation was used for performance estimation and hyperparameter tuning. The final classification was generated by the meta-classifier based on integrated features. Model evaluation involved assessing the overall framework performance using area under the receiver operating characteristic curve (AUC) and analysing feature importance with mean Gini impurity metrics.

### *Patient and public involvement*

Patients and public had no role in study design, data collection and analysis, decision to publish, or manuscript writing.

## RESULTS

### *Endotype and signature identification using omicVAE and K-means clustering*

We first sought to identify endotypes from our sample of 414 patients with KOA. The patient sample was 57% female, with a mean age (±SD) of 65.7 ± 8.7 years and BMI (±SD) of 31 ± 7.1 kg/m$^2$. The majority of subjects had anxiety or depression symptom scores in the normal range, and the majority had painDETECT scores indicating likely nociceptive pain. Mean baseline WOMAC pain score for the sample was 10.1 ± 3.5 points on a 20 total point scale, and baseline WOMAC function score was 34.9 ± 11.9 points on a 68 total point scale (Table 1).

After metabolomics and miRNomics analyses of plasma, synovial fluid and urine, our analytical dataset consisted of 2727 molecular features from 4 domains: 151 plasma metabolites, 421 plasma miRNAs, 930 synovial fluid miRNAs, and 1225 urine miRNAs. We then developed *omicVAE* with K-means clustering (Fig 1), which uncovered 3 clusters of patients using the 4 domains (Fig 2A). The selection of 3 clusters was informed by scree and silhouette plots (Supplementary Fig S1), which visually indicated a significant drop in explained variance (scree plot) and a peak in silhouette score at k = 3 (silhouette plot), suggesting that 3 clusters best represented the underlying data structure. Distribution of most baseline clinical, demographic and anthropometric measures were similar across clusters, except cluster 3 that had a higher proportion of subjects with depression scores in the normal range, and cluster 1 that had a higher proportion of subjects with likely neuropathic-like pain (Table 2).

Significant features associated with each cluster were identified by the intersection of DE and SMD analyses. Distinct signatures consisting of 30, 16, and 24 features for clusters 1 to 3, respectively, were identified (Fig 2B and Supplementary Table S1). Notably, each signature contained features from all 4 domains. In cluster 1, the highest mean value difference was observed for synovial fluid hsa-miR-2053. In cluster 2, synovial fluid hsa-miR-496 exhibited the highest mean value. In cluster 3, plasma hsa-miR-570-3p had the highest mean value. Thus, each cluster represents a group of subjects with a distinct endotype.

### *Endotype feature signatures are enriched for unique pathways*

We next sought to determine if cluster endotype signatures were associated with unique physiological pathways. We first

Table 1

**Summary statistics of 414 knee OA subjects at baseline undergoing total knee arthroplasty**

| | Full sample (N = 414) |
|---|---|
| Sex, n (%) | |
| Female | 232 (56) |
| Male | 182 (44) |
| Age (years) | |
| Mean (SD) | 65.7 (8.4) |
| Median (Q1, Q3) | 66 (60, 71) |
| BMI (kg/m$^2$) | |
| Mean (SD) | 31.3 (7.0) |
| Median (Q1, Q3) | 29.8 (26.3, 35.0) |
| HADS anxiety, n (%) | |
| Normal | 301 (74) |
| Borderline | 54 (13) |
| Case | 50 (12) |
| Missing, n | 9 |
| HADS depression, n (%) | |
| Normal | 324 (79) |
| Borderline | 48 (12) |
| Case | 37 (9) |
| Missing, n | 5 |
| painDETECT, n (%) | |
| Neuropathic-like | 55 (13) |
| Nociceptive | 255 (62) |
| Unclear | 81 (20) |
| Missing | 23 (6) |
| WOMAC pain at pre-TKA baseline | |
| Mean (SD; min, max) | 10.1 (3.5; 2, 20) |
| Median (Q1, Q3) | 10 (8, 12) |
| WOMAC function at pre-TKA baseline | |
| Mean (SD; min, max) | 34.5 (11.9; 0, 67) |
| Median (Q1, Q3) | 35 (26, 42) |
| Missing, n | 2 |

BMI, body mass index; HADS, hospital anxiety and depression scale; Q, quartile; TKA, total knee arthroplasty; WOMAC, Western Ontario and McMaster Universities Arthritis Index.

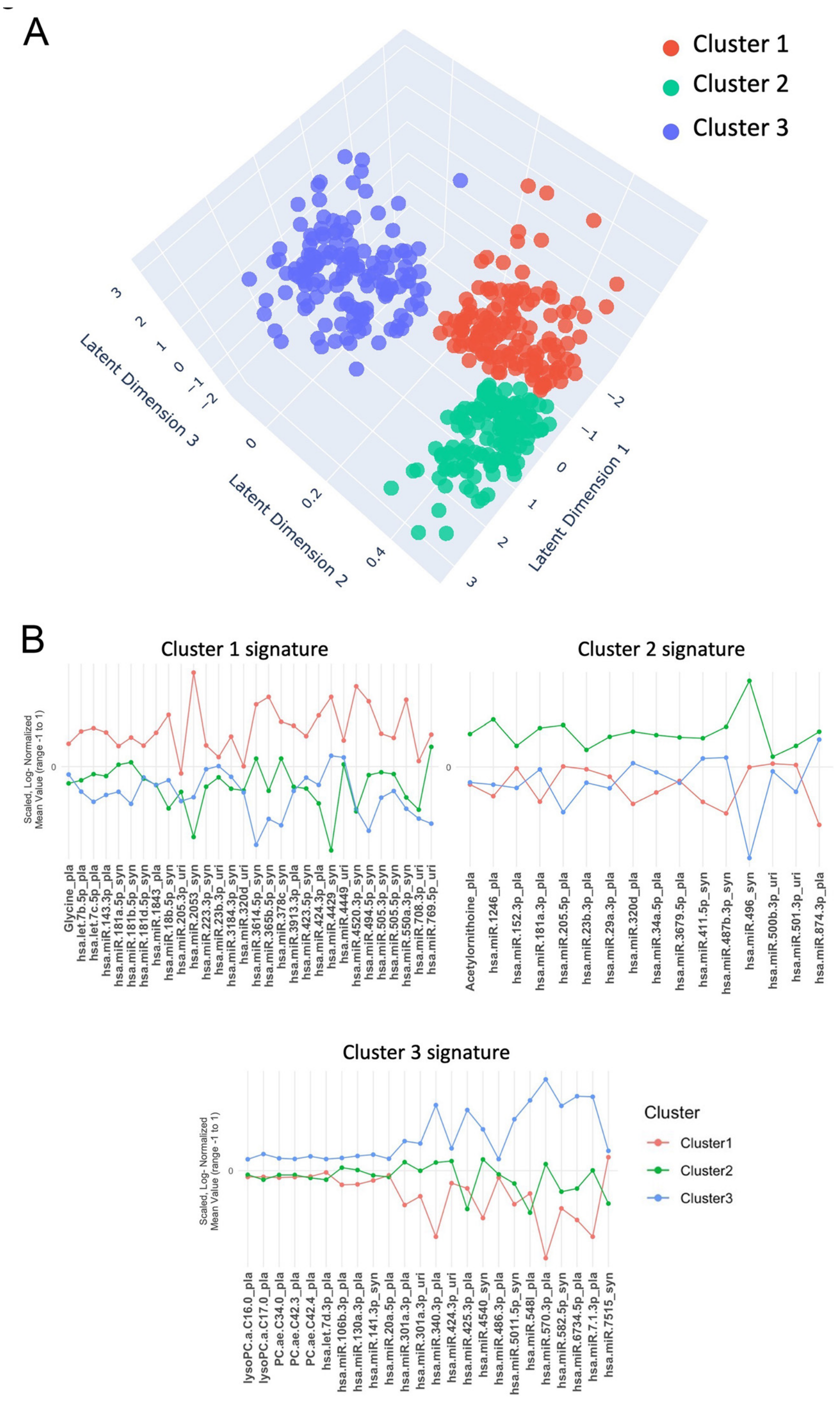


**Figure 2. Integrated analysis of patient clustering and microRNA (miRNA) and metabolite feature signatures.** (A) Three-dimensional illustration (latent dimension 1-3) of the 3 clusters obtained using the variational autoencoder-based deep learning framework. Cluster 1 (red) comprises 146 patients, cluster 2 (green) consists of 138 patients, and cluster 3 (blue) includes 130 patients. (B) Molecular signature profiles for clusters 1, 2, and 3 derived through the intersection of the most significant variables (q < 0.05) identified from both standardised mean differences analysis and differential expression analysis within plasma metabolites and miRNA domains, differentiating each cluster.

identified putative miRNA-gene targets using mirDIP [32], identifying 3257, 2211, and 2319 individual genes targeted by the miRNAs in each of the endotype signatures associated with clusters 1 to 3, respectively. Using these gene set lists, we performed enrichment analysis using pathDIP (Supplementary Tables S2-S4) [33]. For metabolites in each endotype signature, pathway annotations were also identified using pathDIP (Supplementary Tables S5-S7). All pathways were also annotated with categories in pathDIP. For each endotype, individual miRNA-gene targets and metabolites were linked to some common as well as unique pathways. The top unique enriched and annotated pathways linked to miRNA-targeted genes or metabolites, respectively, for each endotype are displayed in a network showing individual pathways and categories (Fig 3A, Supplementary Figs S2-S4).

We next used pathway categories to evaluate physiologically relevant mechanisms linked to each endotype. Each enriched miRNA-derived pathway or annotated metabolite pathway was counted based on its category annotation in pathDIP, scaled and visualised using radar plots (Fig 3B). The cluster 1 endotype signature was most linked to pathway categories associated with development and regeneration, membrane transport, metabolism of various molecules, and the nervous system. The cluster 2 endotype signature was most linked to ageing and cellular community categories. Finally, the cluster 3 endotype signature was most linked to transport and catabolism, signal transduction, sensory system, endocrine system, excretory system, immune system, catabolism, and lipid metabolism categories. Overall, these analyses suggested that features associated with each endotype were uniquely associated with distinct physiological pathways.

Table 2

**Summary statistics of clinical variables within each cluster identified from variational autoencoder machine learning modelling and K-means clustering**

| | Cluster 1 (n = 146) | Cluster 2 (n = 138) | Cluster 3 (n = 130) | *P* value between clusters |
|---|---|---|---|---|
| Sex, n (%) | | | | .9 |
| Female | 84 (58) | 76 (55) | 72 (55) | |
| Male | 62 (42) | 62 (45) | 58 (45) | |
| Age (years) | | | | .37 |
| Mean (SD) | 66.2 (8.2) | 65.4 (8.0) | 65.5 (8.9) | |
| Median (Q1, Q3) | 67 (62, 71) | 65.0 (60.0, 70.8) | 66 (60, 72) | |
| BMI (kg/m$^2$) | | | | .55 |
| Mean (SD) | 30.7 (6.1) | 31.9 (7.7) | 31.4 (7.3) | |
| Median (Q1, Q3) | 29.2 (26.4, 33.8) | 30.0 (26.5, 35.7) | 31.1 (25.3, 34.7) | |
| HADS anxiety, n (%) | | | | .72 |
| Normal | 103 (72) | 98 (73) | 100 (79) | |
| Borderline | 21 (15) | 20 (15) | 13 (10) | |
| Case | 19 (13) | 17 (13) | 14 (11) | |
| Missing, n | 3 | 3 | 3 | |
| HADS depression, n (%) | | | | **.037** |
| Normal | 107 (74) | 105 (78) | 112 (87) | |
| Borderline, | 23 (16) | 19 (14) | 6 (5) | |
| Case, | 15 (10) | 11 (8) | 11 (9) | |
| Missing, n | 1 | 3 | 1 | |
| painDETECT, n (%) | | | | **.011** |
| Neuropathic-like | 30 (21) | 14 (10) | 11 (8) | |
| Nociceptive | 86 (59) | 82 (59) | 87 (67) | |
| Unclear | 27 (18) | 31 (22) | 23 (18) | |
| Missing | 3 (2) | 11 (8) | 9 (7) | |
| WOMAC pain at pre-TKA baseline | | | | .98 |
| Mean (SD) | 10.2 (3.5) | 10.0 (3.2) | 10.0 (3.7) | |
| Median (Q1, Q3) | 10 (8, 12) | 10 (8, 12) | 10.0 (8.0, 12.4) | |
| WOMAC pain at 1 year post-TKA | | | | .34 |
| Mean (SD) | 3.7 (4.0) | 3.7 (3.6) | 3.2 (3.6) | |
| Median (Q1, Q3) | 3.0 (1.0, 5.8) | 3 (1, 6) | 2.0 (0.2, 5.0) | |
| WOMAC pain change (1 year—baseline) | | | | .55 |
| Mean (SD) | −6.5 (4.1) | −6.3 (3.9) | −6.8 (4.1) | |
| Median (Q1, Q3) | −7 (−9, −4) | −6.0 (−9.0, −3.2) | −7.0 (−9.8, −4.0) | |
| WOMAC pain change categorical, n (%) | | | | .48 |
| ≤33.3% | 27 (18) | 33 (24) | 22 (17) | |
| >33.3% | 119 (82) | 105 (76) | 108 (83) | |
| WOMAC function at pre-TKA baseline | | | | .74 |
| Mean (SD) | 34.7 (11.9) | 34.9 (11.4) | 33.8 (12.3) | |
| Median (Q1, Q3) | 36 (26, 41) | 35.0 (26.2, 43.0) | 34 (26, 41) | |
| Missing, n | 1 | 0 | 1 | |
| WOMAC function at 1 year post-TKA | | | | .24 |
| Mean (SD) | 14.8 (13.7) | 14.7 (12.9) | 12.8 (13.0) | |
| Median (Q1, Q3) | 12.0 (4.2, 21.0) | 11.5 (4.0, 20.8) | 9.0 (2.2, 19.0) | |
| WOMAC function change (1 year—baseline) | | | | .76 |
| Mean (SD) | −20.3 (13.4) | −20.0 (13.8) | −20.2 (13.3) | |
| Median (Q1, Q3) | −20 (−30, −11) | −18.4 (−31.0, −10.0) | −19.0 (−28.8, −11.0) | |
| WOMAC function change categorical, n (%) | | | | .24 |
| ≤33.3% | 33 (23) | 32 (23) | 24 (19) | |
| >33.3% | 112 (77) | 106 (77) | 105 (81) | |
| Missing, n | 1 | 0 | 1 | |

Frequency (percentage) are provided for categorical variables, whereas median (quartile 1, quartile 3) values are presented for continuous variables according to patients within each cluster. *P* value was computed using Wilcoxon rank sum test for continuous variables and chi-square or Fisher's test as appropriate for categorical variables. *P* values in bold are considered statistically significant ($P < 0.05$).

BMI, body mass index; HADS, hospital anxiety and depression scale; Q, quartile; TKA, total knee arthroplasty; WOMAC, Western Ontario and McMaster Universities Arthritis Index.

## *Evaluation of classification performance for WOMAC pain and function responses*

To determine the classification performance of the clusters for identifying post-TKA WOMAC pain and function response status, we used an integrative ML framework using 5 domains: plasma metabolites, plasma miRNAs, synovial fluid miRNAs, urine miRNAs, and clinical data (age, sex, BMI, anxiety and depression categories and neuropathic-like pain category; Fig 4A). Subject clusters had similar mean pain and function scores 1 year after TKA, change in scores from baseline to 1 year, as well as pain and function response rates (Table 2). We first conducted DE analysis in each cluster, identifying metabolites with a minimum fold change of 1.1 and miRNAs within each biofluid with a minimum fold change of 1.5 between responders and nonresponders. Of the compared ML approaches, random forests consistently outperformed others in each individual domain (Supplementary Table S8) and was used for the unimodal ML models. Subsequently, we employed 10-fold cross-validation to estimate the AUC and 95% CIs obtained while predicting WOMAC pain and function responses (Supplementary Table S9).

After differential analysis−based feature selection, clusters 1 to 3 retained 250, 87, and 49 features, respectively, for the

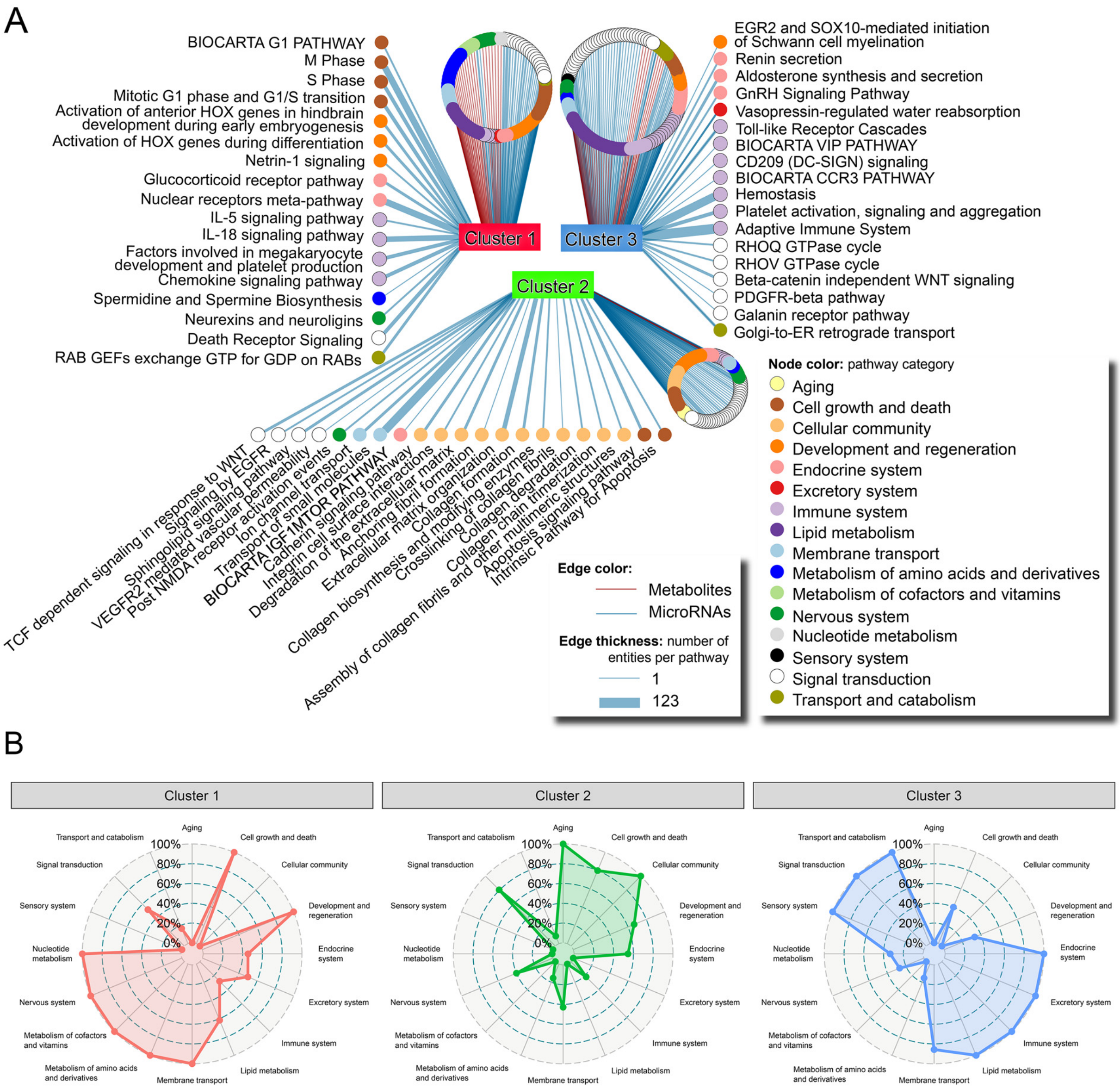


**Figure 3. Cluster endotype signatures are associated with different physiological pathways.** (A) Network depicting unique enriched pathways from microRNA (miRNA)-gene targets and annotated pathways from metabolites associated with individual cluster endotype signatures. Labels show pathways with lowest q value or highest number of annotated genes. (B) Radar plots of pathway categorisations from enriched pathways of miRNA-gene targets and annotated pathways from metabolites indicating categories most associated to individual clusters endotypes.

ML analysis to classify pain response (Supplementary Table S10). Initially, unimodal models were applied to each domain. Within cluster 1, plasma miRNAs demonstrated the highest unimodal performance with an AUC of 0.735. However, the integrative performance, combining all 4 domains in the meta-classifier, notably improved AUC to 0.841 (highlighted in red in the receiver operating characteristic [ROC] plot). For cluster 2, the clinical domain had the highest unimodal AUC of 0.740, whereas the integrative AUC was 0.816. Cluster 3 achieved the highest integrative AUC of 0.879, with urine miRNAs showing the highest unimodal AUC of 0.786 (Fig 2B).

Based on differential analysis−based feature selection to classify function response, clusters 1 to 3 retained 63, 46, and 46 features, respectively, for the ML analysis (Supplementary Table S11). Across clusters 1 to 3, the clinical data domain consistently exhibited the best performance among unimodal domains, with AUCs of 0.702, 0.791, and 0.722, respectively. In terms of integrative performance, clusters 1 to 3 achieved AUCs of 0.786, 0.808, 0.738 (Fig 2C), respectively.

### *Identifying key response classification features in our multimodal ML framework*

To enhance the interpretability of our model we identified the most important features (molecular and clinical) contributing to response classification in each cluster. The top 20 features contributing to WOMAC pain or function response classification are shown in Supplementary Figures S5 and S6, respectively. Each top 20 list consisted of features from all 4 molecular domains, with a notable absence of clinical features; however, all molecular and clinical features inherently played a role in response classification (Supplementary Tables S12 and S13). Interestingly, only 3 miRNAs overlapped in the top 20 important features for WOMAC pain response between clusters 2 and 3, namely synovial fluid hsa-miR-1265 and hsa-mir-642a-3p, and plasma hsa-3942-5p. In addition, only the metabolite glutamine overlapped in the top 20 important feature lists of clusters 2 and 3 for WOMAC function response. Furthermore, within each cluster, the top 20 predictive features were also primarily different for pain and function outcomes, with only synovial fluid hsa-let-

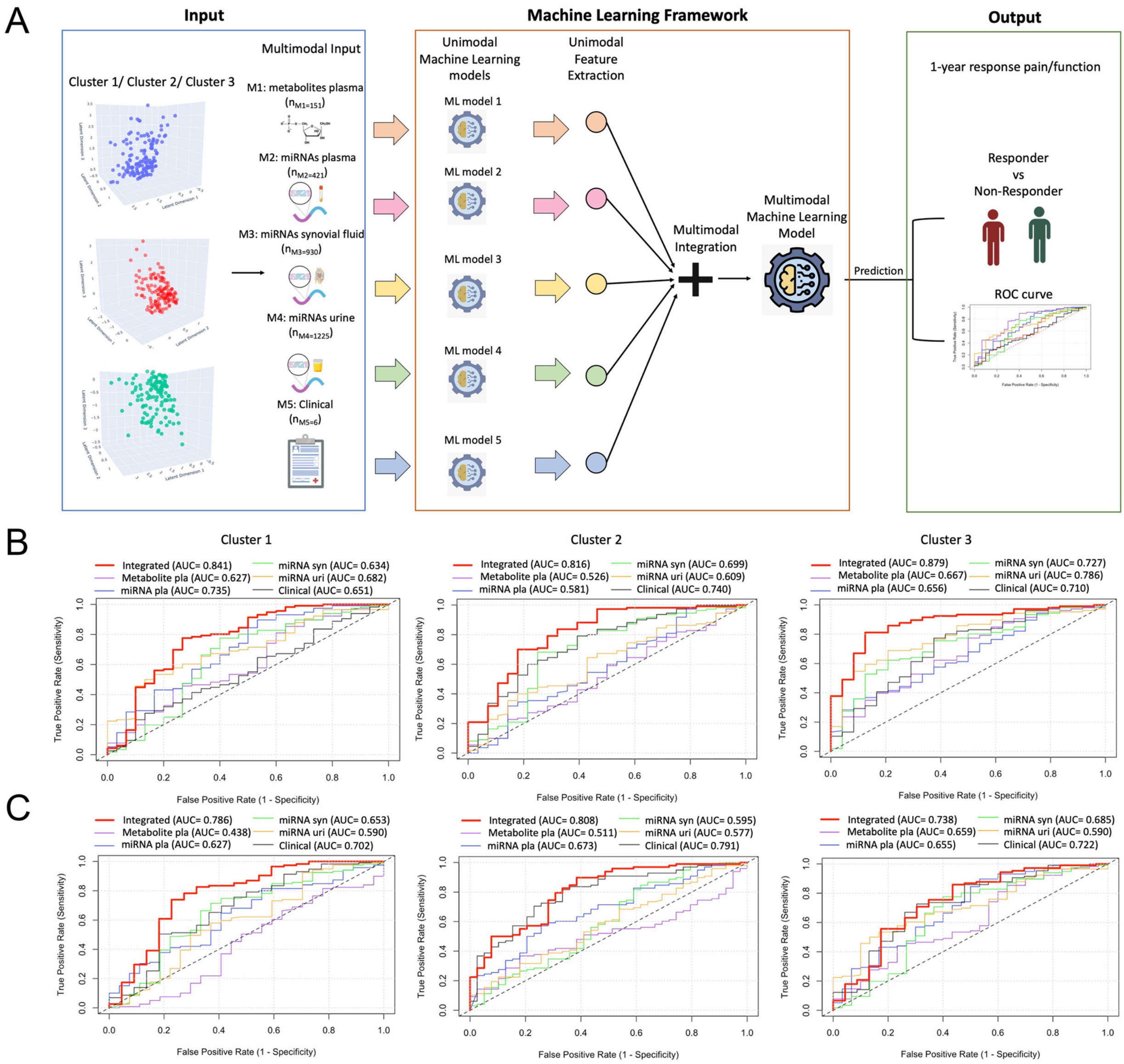

**Figure 4. Machine learning modelling for classifying response to 1 year pain and function within each cluster.** (A) Comprehensive 2-step machine learning (ML) framework wherein, initially, unimodal ML models extract features from each domain, including metabolites from plasma, microRNAs (miRNAs) from plasma, synovial fluid, and urine as well as clinical data. A second-level multimodal machine learning classifier integrates these features to efficiently classify response vs nonresponse at 1 year. (B) Receiver operating characteristic (ROC) plots illustrating the area under the curve (AUC) for individual domains, alongside the integrated AUC (in red) combining all 5 domains to classify Western Ontario and McMaster Universities Arthritis Index (WOMAC) pain response. (C) ROC plots illustrating the AUC for individual domains, alongside the integrated AUC (in red) combining all 5 domains to classify WOMAC function response. Notably, the integrated AUC outperforms individual AUCs in cluster 1, cluster 2, and cluster 3. Created, in part, with BioRender.com.

7f-1-3p in cluster 1 and plasma hsa-miR-3942-5p in cluster 2 showing overlap between lists. Thus, using our integrative approach, the vast majority of the most important features for response classification were unique for each cluster and were also unique between pain and function outcomes within each cluster, with molecular entities primarily driving classification performance. Overall, these findings highlight the diverse molecular features associated with outcome classification in each cluster, emphasising the importance of integrating multiple domains for classification modelling of WOMAC pain and function responses after TKA.

## DISCUSSION

Demographic, anthropometric and clinical characteristics of patients with OA are heterogeneous, influencing outcomes to therapy, including TKA [3,35]. Heterogeneity has also been identified through biofluid data; however, most studies to date have used single biofluids with single molecular type measures to identify endotypes within OA cohorts [13,14]. We developed a novel multimodal deep learning algorithm, *omicVAE*, to cluster a sample of 414 KOA subjects who underwent TKA using preoperative miRNA and metabolite feature sets identified by miRNomics and targeted metabolomics from plasma, synovial fluid, and urine, and uncovered 3 unique cluster endotypes.

*OmicVAE*, the novel VAE architecture described in this study, was specifically designed for integration and clustering of multimodal, multiomic data. Although VAEs have been applied in other omics settings [36−39], *omicVAE* distinguishes itself by its ability to handle heterogeneous data from distinct modalities: metabolomics and miRNomics. A key innovation lies in its architecture, where a single encoder network generates a shared latent space representation from concatenated multimodal inputs, enabling integrative analysis across these domains. To ensure that modality-specific features are preserved while leveraging shared latent information, *omicVAE* incorporates 4 independent decoder networks, each tasked with reconstructing data from a specific input modality. This modular and integrative design makes *omicVAE* uniquely suited for clustering tasks involving complex, multimodal datasets, where both intermodality relationships and domain-specific reconstruction are critical.

To our knowledge, our study is the first to use a subject-matched multi-biofluid, multiomic approach to endotyping patients with KOA. Not only did we uncover 3 unique multiomic-based cluster endotypes, each was linked to unique pathway categories. Despite a similar clinical phenotype, the cluster 1 endotype was primarily linked to metabolic processes and nervous system pathways, the cluster 2 endotype was primarily associated with ageing pathways, and the cluster 3 endotype was primarily linked to immune, endocrine, and lipid metabolism pathways. Overall, the cluster endotypes uncovered are likely to contribute to, or be the result of, distinct mechanisms associated with patients with KOA.

Using this novel approach to cluster endotyping, combined with integrative multimodal ML, we enhanced classification of patient-reported pain and function responses beyond that achieved using clinical measures alone. Importantly, differences in patient-reported pain and function, or response to surgery, were not found simply between patient clusters. Ultimately, significant differences in clinical measures between clusters were only found in the proportions of patients with suspected depression or neuropathic-like pain. In contrast, we found that differences in molecular and clinical measures across patients within each cluster were able to improve classification response. Although the clusters may not differ in pain or function scores, the ML workflow was designed to identify and leverage more subtle, multidimensional patterns such as biomarkers, clinical characteristics, or other high-dimensional features, that may influence outcomes within each cluster. By focusing on nuanced predictive relationships, the ML workflow adds value by identifying individualised predictors of outcomes, even in the absence of broad differences in traditional clinical scores. Consistent with this, we found that molecular entities primarily drove classification of pain and function responses using our integrative modelling; however, all clinical and molecular features included in the model have some contribution to classification modelling performance. Overall, our unique methodological approach reduced OA patient heterogeneity by defining patient clusters that had intracluster molecular differences that enhanced classifying pain and function responses to TKA. As endotypes are further refined and molecular entities best associated with classification are further characterised, it will be important to understand how these response-based signatures may relate to physiological responses after surgery.

The strength of our methodology lies in developing integrative deep learning and ML techniques for efficient multiomic endotyping and response classification in patients with KOA. VAEs offer advantages to integrating multiple domains for clustering compared to traditional approaches by effectively capturing the underlying structure of heterogeneous data through a joint latent representation. Although traditional approaches such as dimensionality reduction or sequential clustering may provide insights, they often suffer from limitations such as difficulty in capturing nonlinear relationships, and inadequate integration of domain-specific characteristics. Unlike standard VAEs [22], we employed 4 separate decoders, enabling domain-specific reconstruction facilitating robust subject clustering, accounting for the inherent uncertainty in the latent space via the VAE's probabilistic nature. Overall, VAEs effectively leverage complementary information from multiple modalities for a more comprehensive characterisation of KOA patient endotypes.

The integration of multiple data domains through our comprehensive 2-step ML framework represents a significant advancement in response modelling for KOA outcomes by combining complementary information inherent in metabolomics, miRNA, and clinical data domains. Utilisation of unimodal ML models in the first step allowed for extraction of domain-specific features that could classify 1-year TKA pain and function responses, while subsequent integration of these features using a Naïve Bayes meta-classifier enhanced classification accuracy. Naïve Bayes classifiers emulate aspects of clinical decision making by probabilistically combining evidence from multiple sources to make classifications [40]. Importantly, our novel framework demonstrated improvements in classification performance compared to unimodal domain-specific models, underscoring the utility of an integrative approach.

Although we included 3 biofluids to integrate miRNA or metabolite features to identify endotypes among patients with KOA, additional endotypes may exist. Incorporating additional omic technologies (eg, proteomics, genomics and methylomics) in the presented framework, as well as comprehensively evaluating omic measures across all biofluids, may further refine endotypes, or uncover additional endotypes to further improve our understanding of KOA and ability to more accurately classify responses to interventions. Future studies should also focus on easily obtained patient biofluids, such as urine and blood, to determine whether the presented approach can show similar endotyping capability and response classification accuracy. As our methodology sought to identify patient clusters only by molecular features, we did not consider whether clinical/sociodemographic/anthropometric/psychosocial variables (such as age, sex, BMI, diet, medications, race, comorbidities, depression and anxiety) may also influence molecular mediators prior to clustering. However, further understanding associations of these factors with OA clusters may provide insights on their relationships with molecular signatures associated with each cluster. In response classification modelling, we only evaluated a subset of clinical/sociodemographic/anthropometric/psychosocial variables associated with KOA and patient outcomes to TKA, but incorporation of additional patient-related factors, and postsurgical therapies such as rehabilitation, alongside endotype data, may also help improve modelling accuracy. Lastly, similar evaluations in additional patient cohorts, such as those with early KOA, with or without disease progression, having other afflicted joints, or evaluating other response measures, would also be of interest moving forward.

Our study is not without limitations. First, plasma and urine were collected within the 3-month period prior to surgery and synovial fluid collection. Although matching fluid collection on the day of surgery would be ideal, as molecular features may fluctuate over time and have some impact on clustering and classification models, this was not logistically feasible for our cohort on day of surgery. In addition, although we extensively validated our integrative ML unimodal models using a 10-time, 10-fold cross-validation, a lack of external validation remains. For external validation to be accomplished, better patient clinical annotations, omic data and biosample sharing practices, and harmonisation are needed [10,11]. Finally, peptide-based endotypes have been identified in previous studies, including low tissue turnover, structural damage and systemic inflammation endotypes [13];

however, we are currently unable to compare these endotypes with those found in our study. Modelling approaches, biomolecules and omic data examined, and cohort sample characteristics can all impact endotype signatures and associated molecular mechanisms, and must be similar for appropriate comparisons.

Overall, using our novel modelling framework, we were able to unravel some heterogeneity of a sample of late-stage surgical patients with KOA and evaluate post-TKA response classification. We anticipate this methodological approach will aid in understanding underlying molecular contributors and pathways to clusters of patients with OA, and define molecular signatures contributing to intervention response. We wish to emphasise that this is the first step in our approach to complex multiomic data integration to understand patient subsets in OA. We provide the framework and methodology for additional molecular, clinical, sociodemographic, anthropometric and psychosocial variables to be included to refine patient clustering and response classification modelling performance to ultimately help in shared patient−clinician decision making for proceeding with selected therapies, including TKA for KOA, improving precision medicine and aiding in subject selection criteria for future clinical trial designs.

## Competing interests

We declare no competing interests.

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## Contributors

JSR, DS, OEG, KH, AS, YRR, AVP, RG and MK conceptualised the study. NNM, K Syed, AVP, YRR, RG and MK supervised patient data and biofluid collection. KP managed biofluid and patient data storage. JSR, DS, OEG, KH, AS, CP, IJ, K Sundararajan, SSL, YRR, AVP, RG and MK processed and curated the data. JSR, DS, OEG, KH, AS, CP, PP, NF, IJ and MK created the methodology and validated the data. DS developed the deep learning and ML algorithms. DS, OEG and KH performed statistical analyses. JSR, DS, CP, IJ and MK created figures and tables. JSR, DS, KH, AS, CP and MK wrote the manuscript. MK supervised and acquired funding for multiomic analysis. All authors had full access to the study data, were involved in manuscript editing, and were responsible for the decision to submit for publication. JSR, DS, OEG, KH, AS and MK are guarantors of the manuscript.

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## Ethics approval

Biofluids and patient data were collected from each patient under informed consent and with approval of the Research Ethics Board at University Health Network under REB approval numbers 07-0383-BE and 14-7592-AE.

## Patient consent for publication

Not Applicable.

## Provenance and peer review

Not comissioned and externally peer-reviewed.

## Data availability statement

Deidentified subject primary miRNA sequencing datasets are available on the Gene Expression Omnibus under accession number GSE222979 [41]. Software code and the dataset of processed miRNA counts, metabolite concentrations and demographic, anthropometric and clinical questionnaire responses used in this study is available at https://github.com/divya031090/DeepLearning_KOA [42].

## Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ard.2025.01.012.

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